PROGRAMMING IN JAVA

Assignment X

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

Which Java package contains networking-related classes?

a. `java.io`

b. `java.util`

c. `java.net`

d. `java.lang`

Correct Answer: c

c. java.net

Detailed Solution:

All core networking classes in Java, such as `URL`, `Socket`, `InetAddress`, and `ServerSocket`, are part of the `java.net` package.

QUESTION 2:

Which class is used to represent an IP address in Java?

a. `IPAddress`

b. `InetAddress`

c. `NetAddress`

d. `HostAddress`

Correct Answer: b

b. InetAddress

Detailed Solution:

The `InetAddress` class represents an Internet Protocol (IP) address and provides methods to obtain host names and IP addresses.

QUESTION 3:

Which method is used to obtain the IP address from an `InetAddress` object?

a. `getAddress()`

b. `getIP()`

c. `getHostAddress()`

d. `getNetworkAddress()`

Correct Answer: c

c. **getHostAddress()**

Detailed Solution:

The `getHostAddress()` method returns the IP address in string format.

QUESTION 4:

Which protocol is used by `Socket` in Java for communication?

a. **UDP**

b. **HTTP**

c. **TCP**

d. **FTP**

Correct Answer: c

c. **TCP**

Detailed Solution:

The `Socket` class uses the TCP protocol, which is connection-oriented and reliable.

QUESTION 5:

Which class is used to create a server-side socket in Java?

a. `Socket`

b. `ClientSocket`

c. `ServerSocket`

d. `NetServer`

Correct Answer: c

c. ServerSocket

Detailed Solution:

The `ServerSocket` class waits for client requests and accepts incoming connections.

QUESTION 6:

Which method converts a `URI` object into a `URL` object?

a. `convertURL()`

b. `getURL()`

c. `toURL()`

d. `openURL()`

Correct Answer: c

c. toURL()

Detailed Solution:

The `toURL()` method converts a `URI` into a `URL` obj.

QUESTION 7:

What value does `getPort()` return if no port is specified in the URL?

a. **0**

b. **80**

c. **-1**

d. **443**

Correct Answer: c

c. **-1**

Detailed Solution:

If a port number is not explicitly mentioned in the URL, `getPort()` returns `-1`.

QUESTION 8:

Which protocol is connectionless?

a. **TCP**

b. **HTTP**

c. **UDP**

d. **SMTP**

Correct Answer: c

c. **UDP**

Detailed Solution:

UDP is a connectionless protocol that sends data as packets without establishing a connection.

QUESTION 9:

Which method is used by a client to establish a connection with a server?

a. `accept()`

b. `connect()`

c. `listen()`

d. `open()`

Correct Answer: b

b. connect()

Detailed Solution:

A client uses the `connect()` method internally when creating a `Socket` to establish a connection with the server.

QUESTION 10:

Which layer of the OSI model does TCP belong to?

a. **Application layer**

b. **Network layer**

c. **Transport layer**

d. **Data link layer**

Correct Answer: c

c. **Transport layer**

Detailed Solution:

TCP operates at the transport layer and provides reliable, ordered, and error-checked data delivery.